- **Date and Time of Collection:** Saturday, 12 March 2022
- **Location:** Tswaing Crater, Gauteng, South Africa

1. Herbaceous Layer Field Data

1.1 Grass Harvest Wet/Dry Weight Inputs

- Sample 10 (10m mark): Mass with bag = 61.82 g | Bag mass = 18.50 g
- Sample 20 (10m replicate): Mass with bag = 92.26 g | Bag mass = 18.50 g
- Sample 30 (20m mark): Mass with bag = 69.62 g | Bag mass = 18.50 g
- Sample 40 (20m replicate): Mass with bag = 57.15 g | Bag mass = 18.50 g

1.2 Herbaceous Layer Species Abundance Frequency Array

- Eragrostis rigidior (Erarig): Rank 1 Count = 23 | Rank 2 Count = 4 | Rank 3 Count = 0 | Total Frequency = 27
- Fabacae unkj (pod): Rank 1 Count = 0 | Rank 2 Count = 0 | Rank 3 Count = 0 | Total Frequency = 0
- Schizachyrium sanguineum (red leaves): Rank 1 Count = 0 | Rank 2 Count = 18 | Rank 3 Count = 2 | Total Frequency = 20
- Brachiaria brizantha (hairy leaf): Rank 1 Count = 0 | Rank 2 Count = 2 | Rank 3 Count = 26 | Total Frequency = 28
- Elongated digitate inflorescences (Eldigfl): Rank 1 Count = 0 | Rank 2 Count = 1 | Rank 3 Count = 0 | Total Frequency = 1
- Semi-digitate inflorescence (semi): Rank 1 Count = 2 | Rank 2 Count = 0 | Rank 3 Count = 0 | Total Frequency = 2
- Hygrostis scabra (SNO): Rank 1 Count = 4 | Rank 2 Count = 0 | Rank 3 Count = 1 | Total Frequency = 5
- Aristida congesta (ARI): Rank 1 Count = 1 | Rank 2 Count = 1 | Rank 3 Count = 0 | Total Frequency = 2
- Gross 2 (gross2): Rank 1 Count = 0 | Rank 2 Count = 1 | Rank 3 Count = 0 | Total Frequency = 1
- Posurv (Pos urv): Rank 1 Count = 0 | Rank 2 Count = 0 | Rank 3 Count = 1 | Total Frequency = 1
- Hetcon (het con): Rank 1 Count = 0 | Rank 2 Count = 1 | Rank 3 Count = 0 | Total Frequency = 1
- Speargrass (spear grass): Rank 1 Count = 0 | Rank 2 Count = 1 | Rank 3 Count = 0 | Total Frequency = 1
- Poly (poly): Rank 1 Count = 0 | Rank 2 Count = 1 | Rank 3 Count = 0 | Total Frequency = 1

1.3 Surface Disturbance and Functional Identifiers (LFA Table Basis)

- Insect Herbivory Rate: 7 out of 7 sampled quadrats show distinct signs.
- Mammalian Herbivory Rate: 0 out of 7 sampled quadrats show active signs.
- Dung Presence Factor: Recorded in 0 out of 7 sampled quadrats.

2. Woody Plant Architecture Attributes

- Net Individuals Sampled (n): 7 trees total over the 20m x 4m belt transect area (80 m^2).
- Leaf Form Groups: 100% compound-leaved stems.
- Active Defense Signatures: 71% feature spines, thorns, hooks, or prickles.
- Parasitic/Predatory Stress: 100% show insect herbivory marks | 0% mammalian browsing.

3. Soil Physical & Hydrological Metrics

3.1 Unbroken Core Buoyancy Metrics (Horizon B1)

- Initial Core Mass: 75.53 g
- Wax-Coated Submerged Weight: 76.24 g
- Suspended Mass In Water: 41.21 g

3.2 Sieve-Separated Matrix Portions

- Total Soil Sample Mass + Bag: 2160.92 g
- Empty Sample Bag Tare Weight: 18.19 g
- Isolated Coarse Gravel Mass (>2mm fraction): 577.46 g

3.3 Sedimentation Hydrometer Testing Arrays

- Hydrometer Readings: 40 seconds (A) = 7.0 | 4 minutes (B) = 8.0 | 2 hours (C) = 6.0
- Slurry Temperature Log: 20.0 °C
- Temperature Correction Constant (TCF): 20.0

3.4 Gravimetric Moisture & Filter Paper Weights

- Crucible Weight (A): 84.39 g
- Crucible + Wet Soil (B): 94.37 g
- Crucible + Dry Soil (C): 94.30 g
- Oven-Dry Target Filter Paper: 0.441 g
- Clean Ziplock Tare Weight: 3.04 g
- Container + Wet Paper Mass: 4.16 g

3.5 Layer Physical Depth Structuring

- Horizon 1 (Layer 1) Layer Thickness (z): 0.39 m
- Horizon 2 (Layer 2) Layer Thickness (z): 0.26 m
- Horizon 3 (Layer 3) Layer Thickness (z): 0.27 m

4. Soil Hydrochemistry & Nutrient Fluxes

4.1 Solution Calibration Constants (Horizon B1)

- Supernatant Reading Temperature: 19.2 °C
- Raw Liquid Electroconductivity (EC): 15 mS m^{-1}
- In-Situ Water Extract pH (pH_H2O): 6.79
- Displaced Ionic Extract pH (pH_1M KCl): 3.70

- Extract Acidity Titrant Used (0.01 M NaOH): Sample = 0.80 ml | Blank = 0.20 ml
- Direct Saturation Extract Factors: EC = 68.0 mS/m | pH_water = 5.39 | pH_1M KCl = 4.68 | Extraction Temp = 19.8 °C

4.2 Nitrogen Incubation Testing Parameters

- Fresh Sample Volumetric Scoop: 5.0 ml
- Incubation Setup Time: 7 days at 30 °C in an anaerobic chamber
- Initial Snapshot State (t0): $N\text{-}NO_3^-$ = 0.0 mg/L | $N\text{-}NH_4^+$ = 0.0 mg/L
- Final Yield State (t1): $N\text{-}NO_3^-$ = 6.0 mg/L | $N\text{-}NH_4^+$ = 1.0 mg/L

4.3 Target Displaced Atomic Cations

- Calcium (Ca^{2+}): 1.307 cmol(+) kg^{-1}
- Magnesium (Mg^{2+}): 0.777 cmol(+) kg^{-1}
- Potassium (K^+): 0.200 cmol(+) kg^{-1}
- Sodium (Na^+): 0.043 cmol(+) kg^{-1}

5. Topographic Catena Matrix Boundaries

- Bottom Slope Plot: 36.0% Sand | 31.6% Clay
- Middle Slope Plot: 66.0% Sand | 17.0% Clay
- Top Ridge Slope Plot: 69.4% Sand | 12.9% Clay

6. Aquatic Channel Hydrodynamics

- Equidistant Point Distance Step (w): 1.31 m
- Channel Flow Cross-Section Depths (cm): [15.0, 19.0, 16.0, 15.0, 17.0, 19.0, 24.0, Edge 29.0]
- Dynamic High Water Head (cm): [15.0, 18.0, 15.0, 16.0, 20.0, 20.0, 24.0, Edge 30.0]
- Dynamic Low Water Head (cm): [12.0, 14.0, 8.0, 15.0, 10.0, 18.0, 23.0, Edge 29.0]